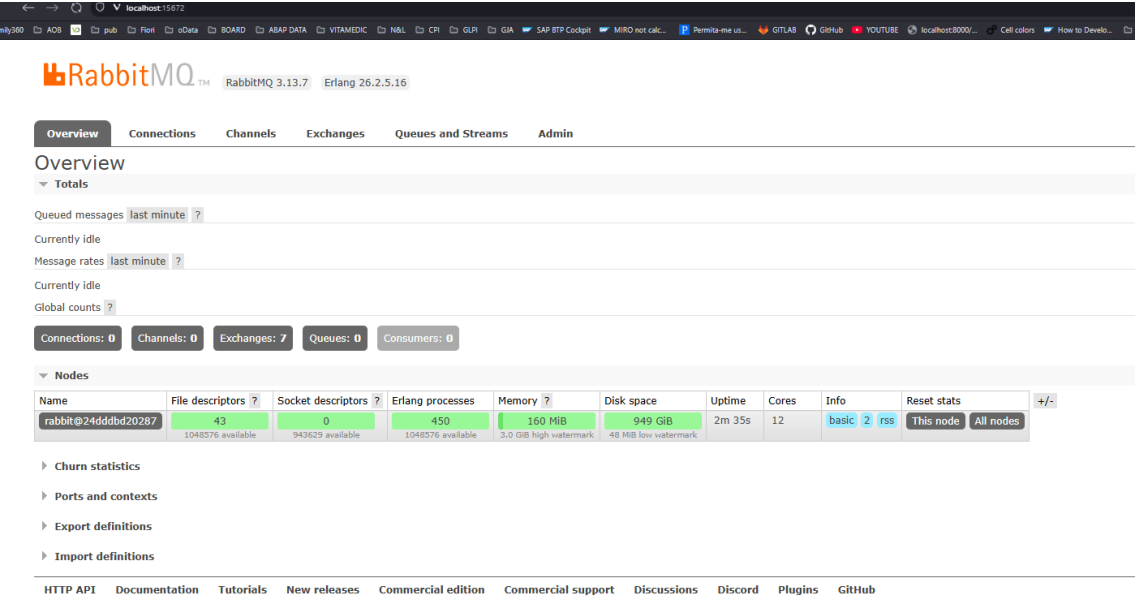
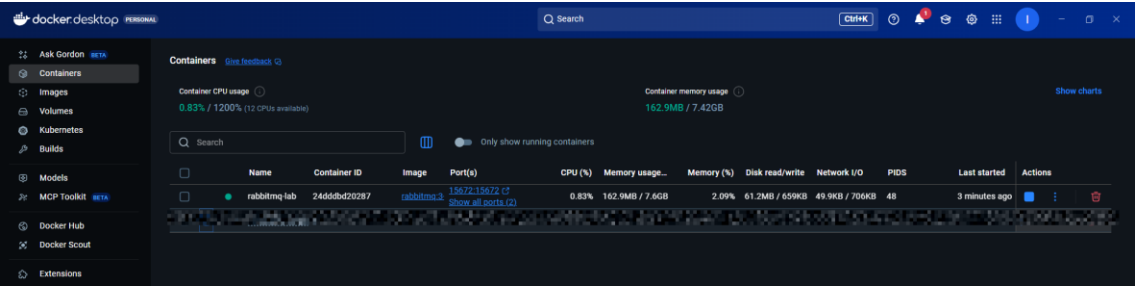


Laboratório 05 - RabbitMQ - Tipos de exchange: Direct e Topic

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1. Subir RabbitMQ com Docker



2. Criar Exchange

a. Vá em Exchanges → Add Name: exchange.direct.lab Type: direct Durability: Durable

Overview Connections Channels **Exchanges** Queues and Streams Admin

Exchanges

▼ All exchanges (8)

Pagination

Page 1 of 1 - Filter: ☐ Regex ?

Virtual host	Name	Type	Features	Message rate in	Message rate out	+/-
/	(AMQP default)	direct	D			
/	amq.direct	direct	D			
/	amq.fanout	fanout	D			
/	amq.headers	headers	D			
/	amq.match	headers	D			
/	amq.rabbitmq.trace	topic	D I			
/	amq.topic	topic	D			
/	exchange.direct.lab	direct	D			

▼ Add a new exchange

b. Pesquise (veja a interrogação ao lado) e explique as seguintes opções:

i. Auto delete

A exchange se apaga automaticamente quando não estiver mais sendo usada (ou seja, quando todas as filas/exchanges que estavam ligadas a ela forem removidas).

ii. Internal

A exchange não pode receber mensagens direto de aplicações. Ela só funciona recebendo mensagens de outras exchanges (exchange → exchange).

3. Criar Filas

a. Crie 2 filas (com Durability = Durable):

- fila.email

● fila.sms

Overview	Connections	Channels	Exchanges	Queues and Streams	Admin
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Queues

▼ All queues (2)

Pagination

Page 1 of 1 - Filter: ☐ Regex ?

Overview					Messages			Message rates			+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack	
/	→ fila.email	classic	D	running	0	0	0				
/	→ fila.sms	classic	D	running	0	0	0				

▼ Add a new queue

b. Qual o significado das opções Durable e Transient em Durability?

Durable

A fila continua existindo mesmo se o RabbitMQ reiniciar (fica salva no disco).

Transient

A fila não é salva. Se o RabbitMQ reiniciar, ela é perdida.

c. Pesquise e explique as opções Auto expire e Message TTL.

Auto expire (x-expire)

Define um tempo para a **fila** expirar.

Se a fila ficar sem uso (ninguém consome/acessa) por esse tempo, ela é **deletada automaticamente**.

Message TTL (x-message-ttl)

Define um tempo de vida para as **mensagens** dentro da fila.

Se passar desse tempo e a mensagem não for consumida, ela é **descartada automaticamente**.

4. Criar Bindings

- a. Entre em exchange.direct.lab → Bindings: To Queue se Routing Key for fila.email email fila.sms sms

 RabbitMQ™

RabbitMQ 3.13.7

Erlang 26.2.5.16

Overview

Connections

Channels

Exchanges

Queues and Streams

Admin

Exchange: exchange.direct.lab

▼ Overview

Message rates

last minute

 ?

Currently idle

Details

Type

Features

Policy

direct

durable: true

▼ Bindings

This exchange

⇓

To	Routing key	Arguments	
fila.email	email		Unbind
fila.sms	sms		Unbind

Add bindings from this exchange

5. Testar mensagens

a. Enviar 1: Routing key: email Payload: {"msg": "Enviar email"}

RabbitMQ 3.13.7 Erlang 26.2.5.16

Overview Connections Channels **Exchanges** Queues and Streams Admin

▼ Bindings

This exchange

↓

To	Routing key	Arguments	
fila.email	email		Unbind
fila.sms	sms		Unbind

Message published.
Close

Add binding from this exchange

To queue: Routing key: Arguments: String

Bind

▼ Publish message

Routing key: email
Headers: ? String
Properties: ?
Payload: {"msg": "Enviar email"}
Payload encoding: String (default)
Publish message

Verificar o resultado e mostrar com um print (gráfico ou get messages):

fila.email → recebe

RabbitMQ 3.13.7 Erlang 26.2.5.16

Overview Connections Channels Exchanges **Queues and Streams** Admin

Queues

▼ All queues (2)

Pagination

Page 1 of 1 - Filter: ☐ Regex ?

Overview					Messages			Message rates				
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack		
/	fila.email	classic	D	running	1	0	1	0.00/s				
/	fila.sms	classic	D	running	0	0	0					

▼ Add a new queue

fila.sms → não recebe

Overview					Messages			Message rates				
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack		
/	fila.email	classic	D	running	1	0	1	0.00/s				
/	fila.sms	classic	D	running	0	0	0					

b. Enviar 2:

Routing key: sms

Payload: Estudando o Message Broker RabbitMQ

The screenshot shows the RabbitMQ web interface for publishing a message. At the top, there's a table with two rows: 'fila.email' with routing key 'email' and 'fila.sms' with routing key 'sms'. Below this, there's a 'Publish message' section. A red box highlights the 'Routing key: sms' and 'Payload: Estudando o Message Broker RabbitMQ' fields. A green arrow points from the 'Publish message' button to a 'Message published.' notification box.

To	Routing key	Arguments
fila.email	email	
fila.sms	sms	

Add binding from this exchange

To queue: Routing key: Arguments: String

Bind

Publish message

Routing key: sms Headers: ? Properties: ? Payload: Estudando o Message Broker RabbitMQ

Payload encoding: String (default)

Publish message

Message published. Close

Verificar o resultado e mostrar com um print:

fila.email → não recebe

fila.sms → recebe

The screenshot shows the RabbitMQ web interface for 'Queues and Streams'. It lists two queues: 'fila.email' and 'fila.sms'. The 'Ready' column for 'fila.sms' is highlighted with a green box and a green arrow.

Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack
/	fila.email	classic	D	running	1	0	1	0.00/s		
/	fila.sms	classic	D	running	1	0	1	0.00/s		

Queue fila.sms

► Overview

► Consumers (0)

► Bindings (2)

► Publish message

▼ Get messages

Warning: getting messages from a queue is a destructive action. ?

Ack Mode:

Encoding: ?

Messages:

[Get Message\(s\)](#)

Message 1

The server reported 0 messages remaining.

Exchange	exchange.direct.lab
Routing Key	sms
Redelivered	0
Properties	delivery_mode: 2 headers:
Payload	37 bytes Encoding: string Estudando o Message Broker RabbitMQ

c. Enviar 3:

Routing key: outra

Verificar o resultado e mostrar com um print (mensagem informativa ou print do gráfico): Nenhuma fila recebe

OverviewConnectionsChannelsExchangesQueues and StreamsAdmin

▼ Bindings

This exchange

⇓

To	Routing key	Arguments	
fila.email	email		Unbind
fila.sms	sms		Unbind

Add binding from this exchange

To queue

:

Routing key:

Arguments:

=

String

▼

Bind

▼ Publish message

Routing key:

outra

Headers: ?

=

String

▼

Properties: ?

=

Payload:

Mussum Ipsum, cacilds vidis litro abertis. Manduma pindureta quium dia nois paga. Praesent malesuada urna nisi, quis volutpat erat hendrerit non. Nam vulputate dapibus. Praesent vel viverra nisi. Mauris aliquet nunc non turpis scelerisque, eget. Nulla id gravida magna, ut semper sapien.

Payload encoding:

String (default)

▼

Publish message

▶ Delete this exchange

Message published, but not routed.

Close

Overview					Messages			Message rates			+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack	
/	fila.email	classic	D	running	1	0	1	0.00/s			
/	fila.sms	classic	D	running	1	0	1	0.00/s	0.00/s	0.00/s	

Topic Exchange

6. Criar Exchange

- a. Criar uma exchange do tipo t pico Name: exchange.topic.lab Type: topic Durability: Durable

Virtual host	Name	Type	Features	Message rate in	Message rate out	+/-
/	(AMQP default)	direct	D			
/	amq.direct	direct	D			
/	amq.fanout	fanout	D			
/	amq.headers	headers	D			
/	amq.match	headers	D			
/	amq.rabbitmq.trace	topic	D I			
/	amq.topic	topic	D			
/	exchange.direct.lab	direct	D	0.00/s	0.00/s	
/	exchange.topic.lab	topic	D			

7. Criar Filas

- a. Crie: fila.pedidos fila.aprovados

Queues

▼ All queues (4)

Page 1 of 1 - Filter: ☐ Regex ?

Overview					Messages			Message rates			+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack	
/	fila.aprovados	classic	D	running	0	0	0				
/	fila.pedidos	classic	D	running	0	0	0				

▼ Add a new queue

8. Bindings

a. Crie os bindings Queue Binding Key

fila.pedidos -> pedido.*

Queue fila.pedidos

► Overview

► Consumers (0)

▼ Bindings (2)

From	Routing key	Arguments	
(Default exchange binding)			
exchange.topic.lab	pedido.*		Unbind

II

fila.aprovados -> *.aprovado

Queue fila.aprovados

► Overview

► Consumers (0)

▼ Bindings (2)

From	Routing key	Arguments	
(Default exchange binding)			
exchange.topic.lab	*.aprovado		Unbind

--

9. Testar mensagens

a. Enviar 1: Routing key: pedido.criado

OverviewConnectionsChannelsExchangesQueues and StreamsAdmin

▼ Bindings

This exchange

⇕

To	Routing key	Arguments	
fila.aprovados	*.aprovado		Unbind
fila.pedidos	pedido.*		Unbind

Message published.

Close

Add binding from this exchange

To queue

*

Routing key

Arguments

=

String

Bind

▼ Publish message

Routing key

pedido.criado

Headers

?

String

Properties

?

Payload

{pedidoId=123}

Payload encoding

String (default)

Publish message

Verificar o resultado e mostrar com um print: fila.pedidos recebe fila.aprovados
não recebe

Overview					Messages			Message rates				+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver	/ get	ack	
/	fila.aprovados	classic	D	running	0	0	0					
/	fila.pedidos	classic	D	running	1	0	1	0.00/s				

b. Enviar 2: Routing key: pedido.aprovado Verificar o resultado e mostrar com um print: fila.pedidos recebe fila.aprovados recebe

To	Routing key	Arguments	
fila.aprovados	*.aprovado		Unbind
fila.pedidos	pedido.*		Unbind

Add binding from this exchange

To queue: *

Routing key:

Arguments: = String

Bind

▼ Publish message

Routing key: pedido.aprovado

Headers: ? = String

Properties: ? =

Payload: {pedidoId=342}

Payload encoding: String (default)

Publish message

▶ Delete this exchange

Message published.
Close

Queue fila.pedidos

- Overview
- Consumers (0)
- Bindings (2)
- Publish message

▼ Get messages

Warning: getting messages from a queue is a destructive action. ?

Ack Mode:

Encoding: ?

Messages:

Get Message(s)

Message 1

The server reported 1 messages remaining.

Exchange	exchange.topic.lab
Routing Key	pedido.criado
Redelivered	0
Properties	delivery_mode: 2 headers:
Payload 14 bytes Encoding: string	{pedidoId=123}

Message 2

The server reported 0 messages remaining.

Exchange	exchange.topic.lab
Routing Key	pedido.aprovado
Redelivered	0
Properties	delivery_mode: 2 headers:
Payload 14 bytes Encoding: string	{pedidoId=342}

- Move messages

c. Enviar 3:

Routing key: pagamento.aprovado

↓

To	Routing key	Arguments	
fila.aprovados	*.aprovado		Unbind
fila.pedidos	pedido.*		Unbind

Message published.
Close

Add binding from this exchange

To queue:

Routing key:

Arguments: = String

Bind

▼ Publish message

Routing key:

Headers: ? = String

Properties: ? =

Payload:

Payload encoding: String (default)

Publish message

Verificar o resultado e mostrar com um print: fila.pedidos não recebe
fila.aprovados recebe

Queue fila.aprovados

► Overview

► Consumers (0)

► Bindings (2)

► Publish message

▼ Get messages

Warning: getting messages from a queue is a destructive action. ?

Ack Mode:

Encoding: ?

Messages:

Get Message(s)

Message 1

The server reported 0 messages remaining.

Exchange	exchange.topic.lab
Routing Key	pagamento.aprovado
Redelivered	0
Properties	delivery_mode: 2 headers:
Payload	{pagamentoId=2332}
18 bytes Encoding: string	

► Move messages

d. Enviar 4 (teste avançado):

Routing key: pedido.cancelado

⇕

To	Routing key	Arguments	
fila.aprovados	*,aprovado		Unbind
fila.pedidos	pedido.*		Unbind

Message published.
Close

Add binding from this exchange

To queue: *

Routing key:

Arguments: = String

Bind

Publish message

Routing key: pedido.cancelado

Headers: ? = String

Properties: ? =

Payload: {pedidoId=3223}

Payload encoding: String (default)

Publish message

Delete this exchange

Verificar o resultado e mostrar com um print: fila.pedidos recebe fila.aprovados
não recebe

Queue fila.pedidos

► Overview

► Consumers (0)

► Bindings (2)

► Publish message

▼ Get messages

Warning: getting messages from a queue is a destructive action. ?

Ack Mode:

Encoding: ?

Messages:

Get Message(s)

Message 1

The server reported 2 messages remaining.

Exchange	exchange.topic.lab
Routing Key	pedido.criado
Redelivered	•
Properties	delivery_mode: 2 headers:
Payload 14 bytes Encoding: string	{pedidoId=123}

Message 2

The server reported 1 messages remaining.

Exchange	exchange.topic.lab
Routing Key	pedido.aprovado
Redelivered	•
Properties	delivery_mode: 2 headers:
Payload 14 bytes Encoding: string	{pedidoId=342}

Message 3

The server reported 0 messages remaining.

Exchange	exchange.topic.lab
Routing Key	pedido.cancelado
Redelivered	•
Properties	delivery_mode: 2 headers:
Payload 15 bytes Encoding: string	{pedidoId=3223}

► Move messages

e. Explique como funcionam os Wildcards (coringas) utilizadas para realizar os bindings nos tópicos.

No binding de filas, você pode usar dois coringas:

*** (asterisco)**

- Substitui **exatamente uma palavra**
- Ex: pedido.* → recebe pedido.criado, pedido.aprovado, pedido.cancelado

(cerquilha)

- Substitui **zero ou mais palavras**
- Ex: pedido.# → recebe pedido.criado, pedido.aprovado, pedido.cancelado, pedido.pagamento.aprovado, etc.

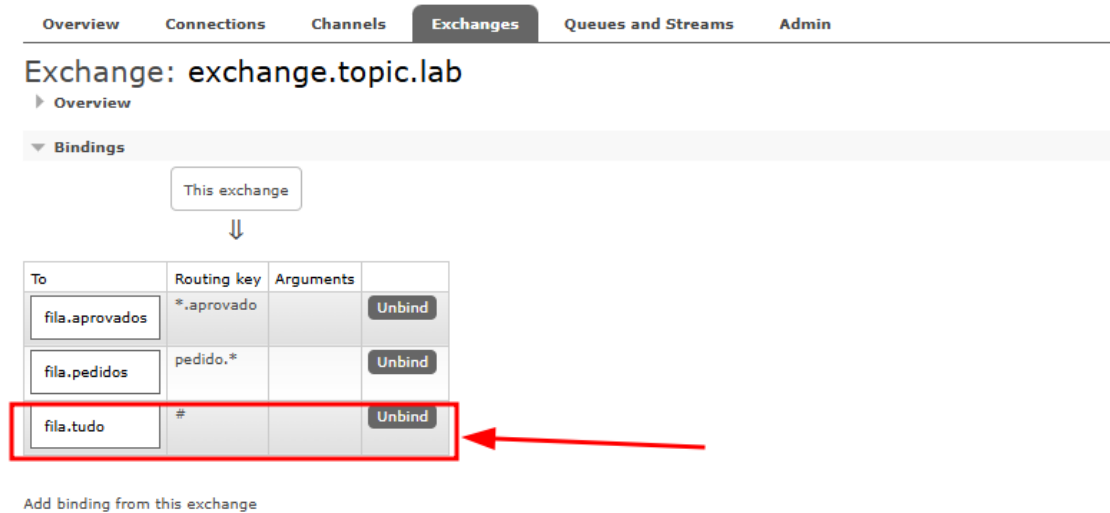
Resumo direto:

- * → uma palavra só
- # → várias palavras (ou nenhuma)

Isso permite filtrar quais mensagens cada fila recebe com base na routing key.

10. Teste complementar

a. Crie uma fila chamada fila.tudo e adicione binding para ela: # → fila.tudo



Overview Connections Channels **Exchanges** Queues and Streams Admin

Exchange: exchange.topic.lab

► Overview

▼ Bindings

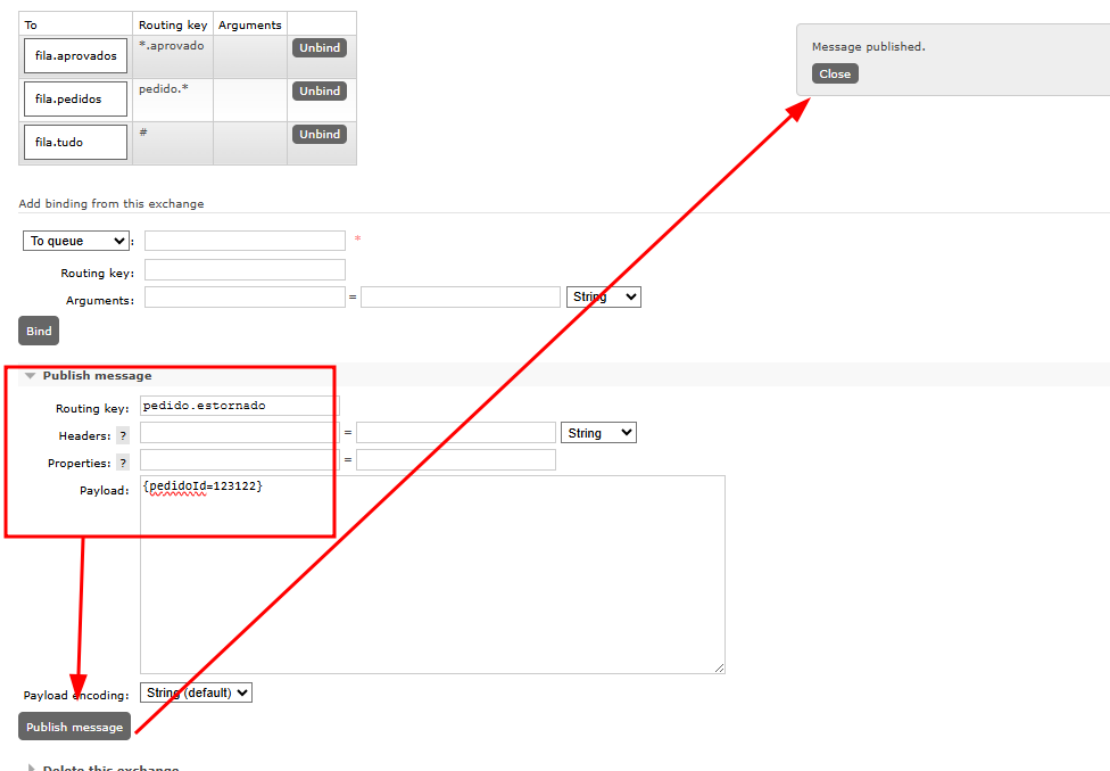
This exchange

⇕

To	Routing key	Arguments	
fila.aprovados	*,aprovado		Unbind
fila.pedidos	pedido,*		Unbind
fila.tudo	#		Unbind

Add binding from this exchange

b. Faça teste - envie mensagens com Routing key diferentes - e, então, explique o que esse binding faz



To	Routing key	Arguments	
fila.aprovados	*,aprovado		Unbind
fila.pedidos	pedido,*		Unbind
fila.tudo	#		Unbind

Add binding from this exchange

To queue: *

Routing key:

Arguments: = String ▼

Bind

▼ Publish message

Routing key: pedido.estornado

Headers: ? = String ▼

Properties: ? =

Payload: {pedidoId=123122}

Payload encoding: String (default) ▼

Publish message

Message published. Close

Overview					Messages			Message rates			+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack	
/	fila.aprovados	classic	D	running	1	0	1		0.00/s	0.00/s	
/ →	fila.pedidos	classic	D	running	4	0	4	0.00/s	0.00/s	0.00/s	
→	fila.tudo	classic	D	running	1	0	1	0.00/s			

O binding com # faz com que a fila receba **todas as mensagens**, independente da routing key.

Por quê?

O # é um wildcard que representa **zero ou mais palavras**, então ele casa com qualquer padrão de routing key.

11. Agora, construa o seguinte cenário para um sistema de e-commerce fictício. Considere o exchange com o nome loja.topico (para mostrar o que foi feito, faça um print da tela do exchange).

Page 1 of 1 - Filter: ☐ Regex ?

Virtual host	Name	Type	Features	Message rate in	Message rate out	+/-
/	(AMQP default)	direct	D			
/	amq.direct	direct	D			
/	amq.fanout	fanout	D			
/	amq.headers	headers	D			
/	amq.match	headers	D			
/	amq.rabbitmq.trace	topic	D I			
/	amq.topic	topic	D			
/	exchange.direct.lab	direct	D	0.00/s	0.00/s	
/	exchange.topic.lab	topic	D	0.00/s	0.00/s	
/	loja.topico	direct	D			

▼ Add a new exchange

- a. Routing keys: pedido.criado pedido.cancelado pagamento.aprovado entrega.enviada
b. Crie filas: fila.pedidos fila.financeiro fila.logistica
c. Bindings esperados: pedidos → pedido.* financeiro → pagamento.* logística → entrega.*

Exchange: loja.topico

► Overview

▼ Bindings

This exchange

⇓

To	Routing key	Arguments	
fila.logistica	entrega.*		Unbind
fila.logistica	pagamento.*		Unbind
fila.pedidos	pedido.*		Unbind

Queues

▼ All queues (7)

Pagination

Page 1 ▼ of 1 - Filter: ☐ Regex ?

Overview					Messages			Message rates			+/-
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver / get	ack	
/	fila.financeiro	classic	D	running	0	0	0				
/	fila.logistica	classic	D	running	0	0	0				
/	fila.pedidos	classic	D	running	4	0	4	0.00/s	0.00/s	0.00/s	

▼ Add a new queue